

End-to-end system for reconstruction of pathological speech using speaker-independent articulatory kinematics

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1. Background

Recent studies have explored using voice conversion (original speaker $\rightarrow$ target speaker) and voice reconstruction (original speaker $\rightarrow$ enhanced original speaker) to improve intelligibility for a variety of speech disorders ranging from speech sound disorders in children [1] to neurodegenerative diseases with secondary dysarthria like Amyotrophic Lateral Sclerosis (ALS) [2]. For neurodegenerative diseases, studies often focus on maintaining speaker identity and generating natural sounding speech [2, 3, 4, 5, 1, 6]. Previous studies have a few key limitations: (1) existing datasets are small and often represent a single disease such that it is hard to make generalizable systems for voice conversion and reconstruction, (2) maintaining speaker identity in the output often involves collecting a large amount of speech data from a single patient (voice banking [7, 8, 9]), which can be burdensome for those with a speech disorder due to symptoms such as fatigue and the progressive nature of many neurological disorders. Some work has attempted to decrease the data size requirements using deep learning models, style transformation techniques, and data augmentation [4, 10, 11, 1, 5].

All existing voice conversion and reconstruction techniques for pathological speech rely on acoustic input, either in the form of waveforms or spectrograms. However, an audio signal is highly variable, and certain acoustic properties of speech are not shared across speakers (i.e., fundamental frequency, intonation). Many conversion techniques have had to first match speakers by gender and age to achieve better performance [12]. More recent work has found that style transformations can generalize across age and between disorders [1].

Another potential way to approach this problem is to project waveforms into the articulatory space. This articulatory space is assumed to be shared across speakers due to the anatomy of the human vocal tract. The vocal tract has a predictable shape and predictable variations (e.g., fundamental frequency), which has led some researchers to conclude that each individual's personal articulatory space is a linear transformation away from a canonical articulatory space [13]. Ground truth articulatory kinematics (measurements of articulator movements over time) can be gathered with methods like electromagnetic articulography (EMA). EMA tracks articulatory movement on the mid-sagittal plane using sensors placed on key articulators (upper lip, lower lip, upper incisor, lower incisor, tongue tip, tongue blade, tongue dorsum; Figure 1) [14]. The resulting output has low dimensionality, consisting of 12 XY measurements of the articulators across time that are often combined with a few key source (voicing) features such as pitch and periodicity to maintain voicing information. However, it is difficult to collect EMA measurements from a large sample size. This has resulted in

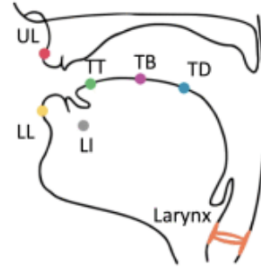


Figure 1: *Electromagnetic Articulography (EMA) sensor placement* [21]

the development of Acoustic-to-Articulatory Inversion (AAI) models that predict these direct articulatory kinematics (EMA) or geometric transformations of these kinematics (articulatory trajectories, TVs) from a waveform [15, 16, 17, 18, 19]. Some systems also resynthesize speech from the predicted articulatory kinematics, which allows for a more interpretable way to manipulate the speech signal [20, 21].

Articulatory measures are also mostly separate from speaker identity. While there is variation in exact articulatory trajectories across speakers, this variability is minimal in comparison to the variability in an acoustic signal for the same vocalization. Additionally, the variability may be accounted for through normalization techniques, transformation into a canonical speaker articulatory space with affine projections, or training of AAI models with multiple speakers to learn more general measurements [13, 21, 22]. Beyond this, many AAI systems model speaker identity independently and recombine learned speaker identity embeddings with the articulatory kinematics at the resynthesis stage, such that articulatory sequence can be edited with less effect on salient speaker identity features [23, 21, 20, 24].

The current state-of-the-art AAI systems have high resynthesis performance on healthy speakers [21], and have shown success on few-shot generalization across speakers and languages [19, 16, 21, 25]. Notably, while no work has used AAI for conversion or reconstruction of pathological speech, [25] presents a zero-shot, real-time voice conversion system for healthy speakers. These capabilities are highly desirable for assistive speech technologies. Some other studies have used predicted articulatory kinematics to help classify healthy speakers from speakers with dysarthria [18] and subtype errors for speakers with speech disorders [26, 27, 15].

2. Research statement

For patients with neurological speech disorders, difficulties producing intelligible speech can have profound effects on their quality of life. A large body of work has attempted to develop assistive speech technology that can improve intelligibility of impaired speech, but variability in the acoustics of a speech signal has made it difficult to maintain speaker identity and naturalness in generated speech without hours of data from patients. As a result, existing acoustic-based voice conversion and reconstruction techniques for pathological speech have focused on reducing data needs with deep learning, data augmentation, and style transformation [4, 10, 11, 1, 5]. These models are still limited in terms of generalizability and by the fact that speaker identity features are heavily entangled in the acoustic space.

Recent work in Acoustic-to-Articulatory inversion (AAI) has demonstrated that speaker identity and speech content can be successfully disentangled projecting speech into an articulatory space while also learning a separate speaker identity embedding [21]. AAI models have had great zero-shot success on downstream tasks of speech editing and voice conversion for healthy speakers [21, 25]. However, these models have not yet been used for improving intelligibility of pathological speech. Given that AAI models have highly desirable properties such as separation of speaker identity and speech content and zero-shot generalization capabilities, I propose to develop an end-to-end AAI system with a novel articulatory conversion/reconstruction module that learns mappings between healthy and disordered articulatory kinematics to improve intelligibility of disordered speech while maintaining speaker identity and naturalness. In cases where speech is too degraded to reconstruct or a disorder is progressive, this method may also allow speaker identity information to be banked instead of voice samples, which would decrease the burden on the patient to produce speech.

3. Proposed design

The voice reconstruction system will start with a pre-trained acoustic-to-articulatory inversion (AAI) model that extracts articulatory kinematics and speaker identity from a waveform. The articulatory kinematics will be sent through a reconstruction module that reconstructs the sample to have more similarity with healthy articulatory kinematics, and then the edited articulatory measure will be combined with the previously extracted speaker identity embedding to resynthesize speech. Some potential designs and training pipelines are shown in Figure 2, 3, 4.

The goal of this system is to improve the intelligibility of pathological speech while maintaining the identity of the speaker. Given the existence of acoustic conversion models for pathological speech and the recent improvements in AAI for healthy speakers, training the articulatory conversion/reconstruction module will be the novel contribution of this project. Following previous studies, the automatic metrics of interest will be word error rate (WER) as a proxy for intelligibility [4, 20, 10, 11, 2], speaker embedding similarity [1, 25] to quantify how well the model maintained speaker identity in the output waveform, and UTMOS, a machine-evaluated mean opinion score (MOS) to quantify naturalness [25]. Acoustic speech conversion and reconstruction methods for dysarthric speech will be used as baseline performance. These metrics are not perfect as intelligibility proxy metrics will not reveal whether pathological speech features have been fully removed from the sample and automatic measures in general may not align with hu-

man perception of naturalness, speaker identity, and intelligibility. Also, measures like WER rely on an external ASR system, and aspects of that system (e.g., inclusion of a language model) could have implications for how well the transcriptions model intelligibility.

Throughout development, we may explore different evaluation paradigms. There are some novel evaluation metrics presented in [22], and it will be important to include some form of human evaluation of naturalness and identity preservation. Also, there will be a need to find or create out-of-sample test sets to measure the system’s generalizability across demographics, speech disorder types, and disorder severity. We may also consider whether there is a correlation between the end-to-end metrics (WER, speaker similarity, UTMOS) and intermediate metrics from the articulatory conversion/reconstruction module as this would allow us to justify using these metrics to quantify how well the model removes pathological speech features and to analyze fine-grained articulatory differences from automatically generated EMA (especially when EMA is generated from single speaker models) [22, 17, 13, 27, 26].

Our initial end-to-end system will use the SPARC [21] AAI model for automatically generating EMA measurements and resynthesizing the waveform. A pre-trained version of this model is available for immediate use on GitHub [21]. The baseline articulatory conversion/reconstruction module will learn to edit EMA measurements from pathological speech using target samples from healthy speakers. The original and target EMA measurements will be aligned with dynamic time warping (DTW), and a simple linear model will be trained to minimize differences between the two signals. The primary metric will be correlation between the original and target EMA measurements. We will also consider if dynamic time warping mel cepstral distortion (DTW-MCD) is a useful metric for training this model [19].

Once a baseline of performance is established and compared to acoustic baselines, we will work on improving performance through changes to the architecture and development of a novel loss function.

While training the articulatory conversion/reconstruction model will be that focus of the project, there may be ways to boost performance by selecting the right AAI models. Many recent studies on AAI have explored a variety of pre-training and fine-tuning strategies using self-supervised speech representations to improve the predicted articulatory kinematics [21, 22, 18, 17, 13, 20, 15, 16, 19]. WavLM [28] seemed to be the primary choice of model [21, 22, 20, 17]. However, the studies showed varied results in terms of the best layer to use (single layer vs. weighted sum, early vs. late layers) [?, 20, 22]. The studies also varied in terms of whether the base SSL model remained frozen, had all the weights fine-tuned, or was fine-tuned with parameter efficient fine-tuning like Low Rank Adaptations (LoRA, [29]). Also, there are various training strategies and model architectures used for AAI models.

One major area of debate is whether an AAI model can be trained with a canonical speaker (single speaker) versus multiple speakers. [13] argues that all individual articulatory spaces are a linear transformation away from a canonical space, such that a canonical speaker can be used to train an AAI model, but [22] argues that multi-speaker AAI always outperforms single-speaker AAI. This could have an impact on voice conversion and reconstruction in the articulatory space as using a canonical speaker could bias the resulting predictions in unexpected ways.

There are also varying opinions on the type of articulatory kinematics to use and which source features to include.

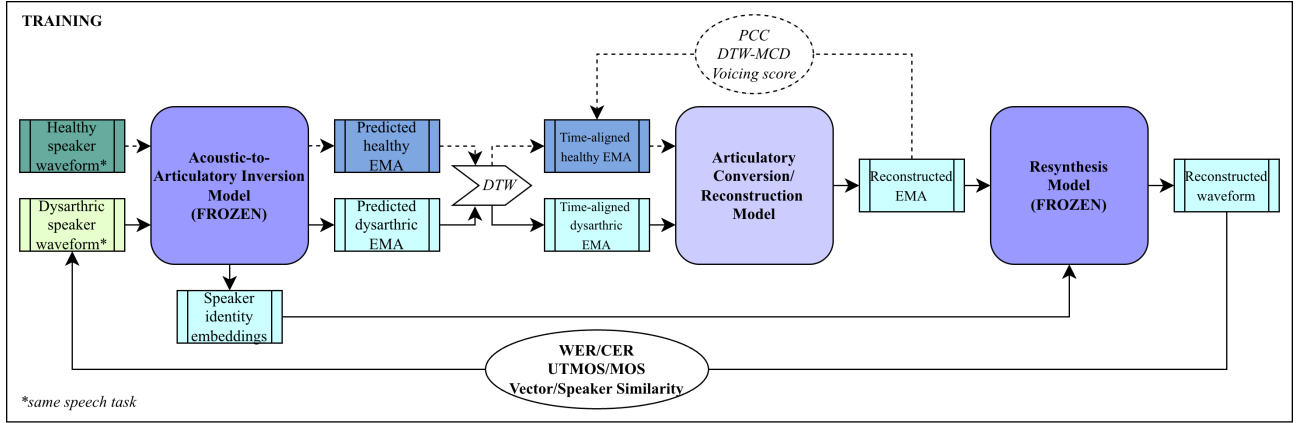


Figure 2: Potential end-to-end training pipeline for voice conversion/reconstruction of pathological speech.

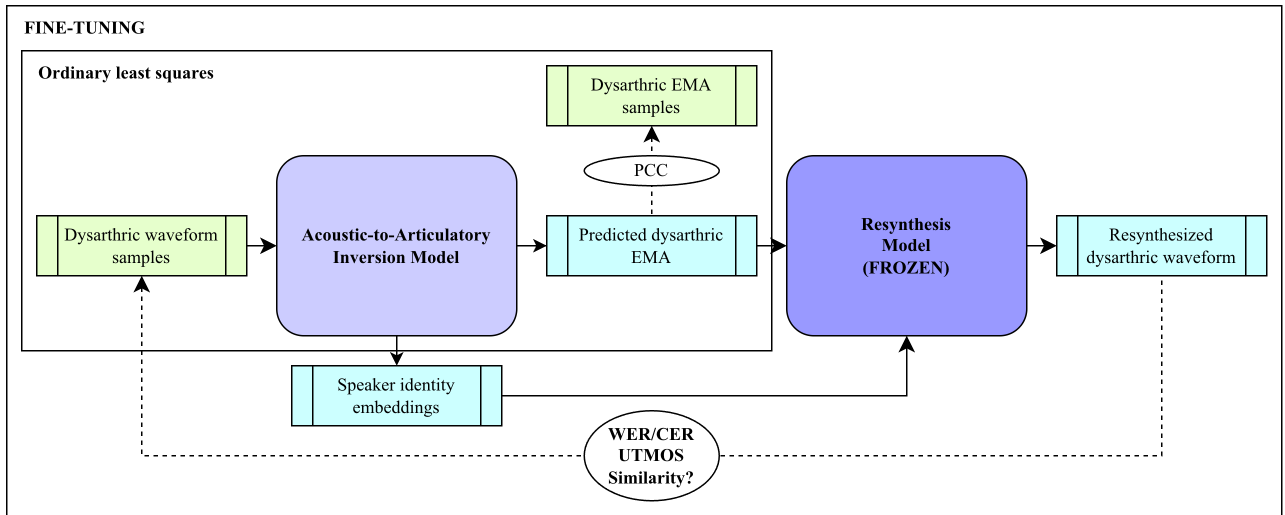


Figure 3: Potential pipeline for finetuning an Acoustic-to-Articulatory Inversion (AAI) model with dysarthric speech.

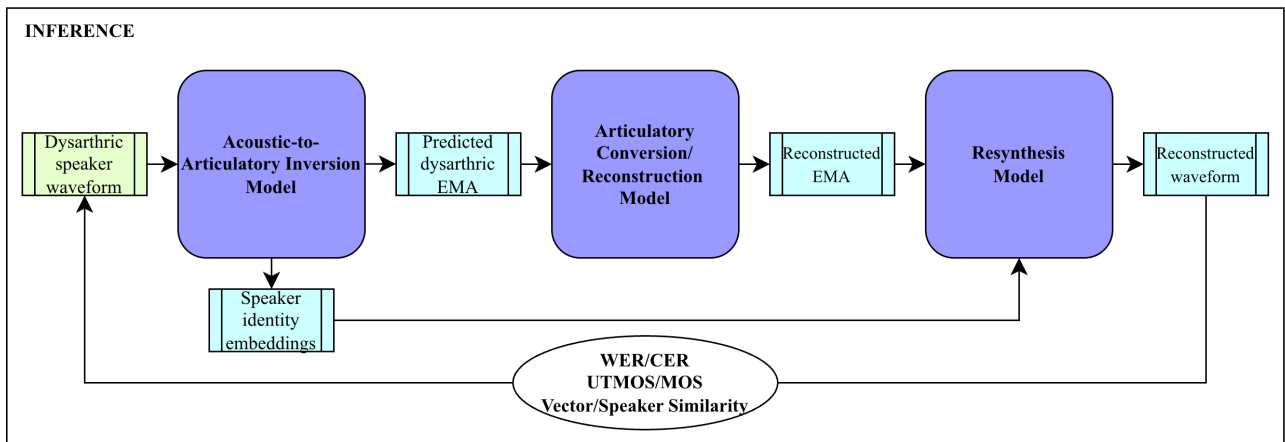


Figure 4: Inference pipeline for evaluating voice conversion/reconstruction of pathological speech.

Some work predicts EMA measurements directly [13, 21, 16] while others calculate articulatory trajectories (TVs) from the EMA measurements [15, 30, 31, 32, 19]. Each study has slightly unique calculations for TVs (e.g., velocity and acceleration [18]) and the number of TVs calculated can vary (6 vs. 9). In terms of source features, a combination of aperiodicity/periodicity [26, 30, 23], pitch (F0/logarithmic F0) [26, 20, 30, 23], and energy/loudness [21, 20] are used to model voicing, but there is no agreement on which combination to use. A more recent study also concluded that nasalance can improve AAI models, but this is hard to estimate without specialized recording setups [24]. Using the right set of articulatory kinematics and source features may be important for training the articulatory conversion/reconstruction model, so this will need to be explored.

It may also be possible to improve AAI performance for dysarthric speech by fine-tuning existing models with ground-truth EMA measurements from speakers with dysarthria. A few public datasets exist (TORGO, HPRC, etc.), and the Speech Disorders and Technology lab at UT Austin has EMA samples from patients with ALS that could be used for this task.

One final consideration is that each AAI system models speaker identity differently. These methods should be carefully studied to ensure that the modeling method makes sense for the given task.

4. Data needs

We will need a large dataset of speech samples from both healthy speakers and speakers with a diagnosed speech disorder to train a conversion or reconstruction module from scratch. We are requesting access to vowel prolongation, AMR, SMR, sentence repetition, reading passage, and picture description samples from the Mayo Clinic speech dataset, along with their associated severity scores, diagnosis-related records, and any additional demographics that are allowed to be shared under the IRB and HIPAA. This may be supplemented with speech samples from the Speech Disorders and Technology lab that can serve as an out-of-sample test dataset.

We may also use EMA measurements from public datasets such as the TORGO database [33] along with EMA measurements collected by the Speech Disorders and Technology lab to fine-tune the base AAI model.

5. Potential limitations and concerns

In cases where speech is highly degraded, voice reconstruction is highly to fail. A situation like this would require an entirely different pipeline. A potential option is to create a text-to-articulatory gesture module. A simple but likely inefficient method would be to use text-to-speech (TTS) to create a speech waveform and then use AAI to convert this to an articulatory gesture, followed by combining this with a previously extracted speaker identity embedding. However, it is unclear how well speaker embeddings will work in this situation due to loss of speaker identity information due to disorder. The prime case for this alternate method would be one wherein a speaker has saved speaker embeddings from mild severity recordings.

It is also important to consider what aspects of prosody are captured in the articulatory kinematics and source features. The raw waveform contains all the necessary information but is highly complex. In converting to the articulatory space, certain features may be lost. This may be especially true if the selected source features or extracted speaker identity embeddings

contain inadequate information. Respiration and phonation issues may impact speaker identity extraction and may have to be accounted for separately. As mentioned, severe cases will highlight these issues the most.

In terms of training, there is a lack of ground-truth EMA data for fine-tuning the base AAI model, meaning fine-tuning may not be helpful for this step. In this case, the predicted EMA measurements will only be as good as the AAI model.

Additionally, there are concerns with the safety of voice conversion and reconstruction. While any method we develop could offer quality of life improvements to those with speech disorders, it will also contribute to developing systems that can make one speaker sound like another, which brings up questions of privacy and authentication using voice [34].

6. References

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